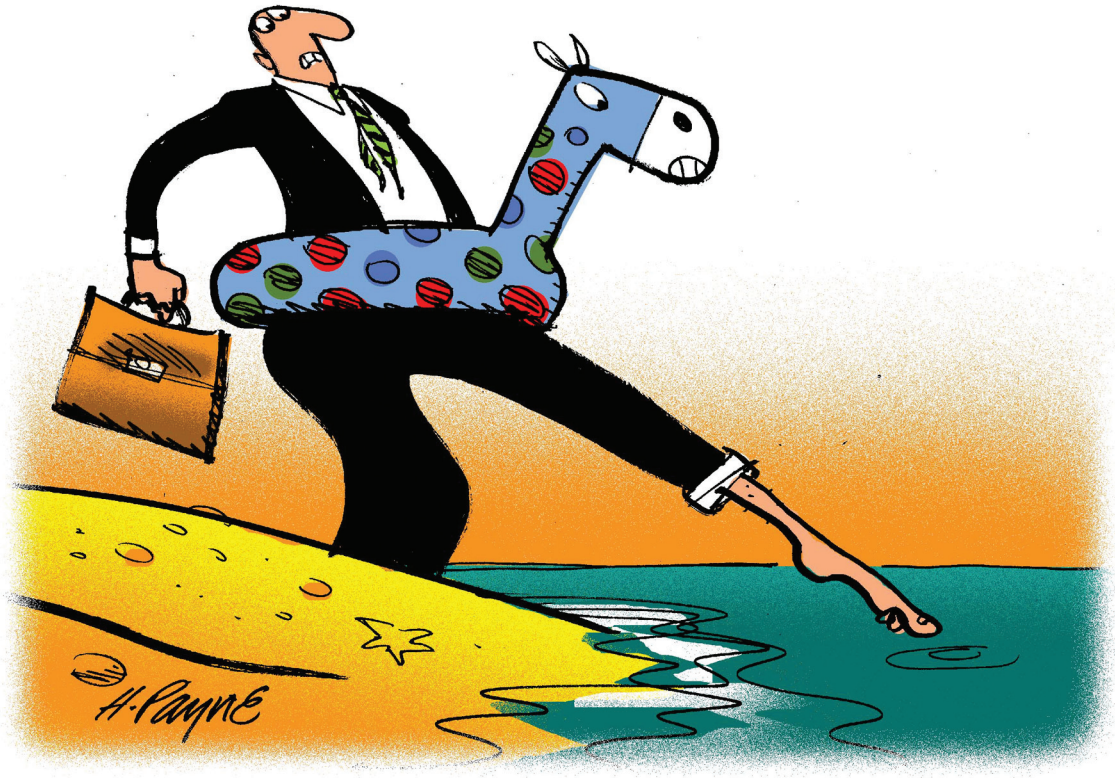
GE's Ann Klee Brings Good Things to Light

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The Disappearing Distinction

The notion of geographical attainment areas is becoming ever less helpful to improving air quality under the Clean Air Act. This change is making the law's State Implementation Plan process less effective. A new methodology is needed

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At the heart of our nation's air quality management system lies the pivotal difference between "attainment" and "nonattainment" areas, generally rural precincts in the first case and population centers in the latter (but not always). Billions of dollars and countless hours are spent on this determination and the economic and public health consequences that flow from it. The difference between attainment and nonattainment, however, is vanishing.

There are three reasons for this disappearing distinction. First, the Clean Air Act's National Ambient Air Quality Standards, reflecting what is needed to protect public health, are closing in on background levels of atmospheric loadings as the NAAQS are revised periodically downwards and emissions from other parts of the world increase. Second, more and more emissions are coming from outside nonattainment areas, or are controlled outside nonattainment areas, making management within the urban area where the people live less effective. And third, energy advancements are making it difficult to geographically link benefits in the form of health improvements with the locus of the reductions, as production moves away from population centers and more distributed alternative energy sources are utilized.

As the distinction vanishes, so is its helpfulness. The disappearance is making the chief statutory method for achieving attainment, the State Implementation Plan process, which implement strategies to achieve the NAAQS in nonattainment areas while taking into account economic considerations, even less effective at improving air quality.

Air quality standards are the foundation of the Clean Air Act. Once a particular NAAQS is promulgated or revised, say for ozone or carbon monoxide, EPA evaluates monitoring data and works with states to make attainment determinations. If monitoring data show levels above the NAAQS, the geographical area is generally determined to be in nonattainment. Otherwise the area is generally considered to be in attainment.

The difference is significant. Nonattainment areas must completely offset emissions from new major stationary sources. Plants in nonattainment areas are also required to install more stringent controls, such as Lowest Achievable Emissions Reduction technology. Nonattainment areas are also subject to a myriad of other requirements such as SIP development, inspection and maintenance programs, and transportation conformity requirements. Just the administrative requirements alone can translate into millions of dollars in expense to taxpayers, states, local governments, and businesses.

One of the reasons for such a significant difference between the requirements for nonattainment areas versus attainment areas is the historic difference in pollution levels between such areas. Now the difference between pollution levels is much less. With this disappearance in the magnitude of pollution levels between attainment and nonat-

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tainment areas, the differing treatment appears to be becoming less helpful in advancing air quality goals.

The NAAQS and background levels are closing in on each other. The previous background level for ozone before its recent revision, for example, was estimated to be around 35 parts per billion. Now background estimates have jumped to approximately 50 ppb. Background pollution is generally defined as the ozone concentration that would occur in the United States in the absence of anthropogenic emissions generated outside North America (i.e., pollution blowing in from other countries outside North America, biogenic emissions, fires, lightning, etc.). On the other side of the equation, the ozone NAAQS level was at 85 ppb. Now the NAAQS level has been revised downward to 75 ppb — with a recommendation to drop it even further to perhaps 60–70 ppb. In other words, the difference between background and the NAAQS used to be 50 ppb. Now the difference is 25 ppb. This difference will continue to decrease to potentially 10 ppb or less as foreign emissions increase in the near term and the NAAQS is periodically lowered.

Also, further diminishing this difference between the ozone NAAQS and background levels is the projected impact of the "climate change penalty." Scientists using empirical data have long shown that ozone is formed more at higher temperatures. If the earth is warming, more ozone will be expected to form. Several studies conducted by EPA, the National Oceanic and Atmospheric Administration, Harvard University, and others are projecting a significant impact of climate change on future U.S. ozone levels. This impact is known as the "climate change penalty." Based on recent research, the ozone climate change penalty might increase currently ozone levels by as much as 15 ppb over the next 40 years. What does this mean? Essentially this means that, holding everything else constant, a background ozone level of 50 ppb in 2012 could become a background level of 65 ppb in 2050 due to the climate change penalty alone.

The difference between background and the NAAQS will continue to decline. The smaller the difference between background and the NAAQS, the less difference (of course) between attainment and nonattainment. And when the difference between living in an attainment area and living in a nonattainment area is as small as 10 ppb, spending billions of dollars to reduce that difference to zero becomes less meaningful, more artificial, and less cost-effective.

The fact of the matter is, as more emissions are coming from outside nonattainment areas or are controlled outside nonattainment areas, the corresponding effectiveness of nonattainment controls becomes less. Examples of emissions sources that impact nonattainment areas but are outside their control are international pollutant transport, interstate pollutant transport, federally preempted mobile sources (i.e. cars, airplanes, ships), and intrastate pollutant transport from power plants, oil and gas sites, and other industries in attainment areas.

Forty-two years ago when the Clean Air Act was written emissions from foreign countries were relatively small and the phenomenon of transcontinental pollutant transport had yet to be discovered. Our understanding of the world has since changed and the world has grown up around us. Just recently NOAA published a report that found that Asian pollution contributed up to 20 percent of the total ozone in the western United States. The study also found that, but for Asian emissions, 53 percent of the ozone standard violations in the West would not have occurred.

In 2010, the "Hemispheric Transport of Air Pollutant Report" for the UN Convention on Long-Range Transboundary Air Pollution found significant and growing impacts from foreign pollution. The report found that the "sum of the health impacts of transported pollution in downwind foreign regions can be larger than the health impacts of emissions in the source region itself." The report also found that as "public health–based air quality standards continue to be tightened based on new health effects research, the contribution of intercontinental transport to exceedances of such standards will continue to increase." The UN report even found that changes in emissions outside North American might be more important than changes made within the continent: "For North America ground-level [ozone] concentrations[,] changes in emissions of [ozone] precursors outside the region may be as important as changes within the region." The report even found that measures at some locations on the West Coast exceeded the NAAQS before even entering the United States.

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The UN came to a bold conclusion. The committee said that "with changing global future emissions, it is likely that over the next 20 to 40 years it will become even more difficult for nations to meet their own environmental policy objectives without international cooperation to address transboundary and intercontinental flows of air pollution."

The Environmental Protection Agency has also participated in foreign pollution studies. In fact, one such study concluded that foreign pollution would thwart U.S. efforts to address pollution within the United States. The study found with regard to a report from the Intergovernmental Panel on Climate Change, "Although fossil fuel emissions in the U.S. for [nitrogen oxides, carbon monoxide, and volatile organic compounds] decline in the IPCC A1 2030 scenario relative to 1995 by 27 percent, 45 percent, and 30 percent, respectively, we find that the number of grid-square summer days over the U.S. with [ozone over] 70 ppb increases relative to 1995. Efforts to improve U.S. air quality are thus thwarted by the rise in global background [ozone] levels due to increased emissions outside U.S. borders."

In 2009, Harvard University looked at pollution from Mexico and Canada. The study concluded that pollution from these two neighbors enhances ozone concentrations in the United states by an average of 3–4 ppb in the summer and can increase ozone concentrations "much higher in downwind regions of the Northeast and Southwest." Foreign pollution affects attainment areas and nonattainment areas alike. As pollution impacts from other countries have increased, the ability of states and nonattainment areas to address the pollution in the air above their region has correspondingly decreased.

One last study worth noting was performed by the National Research Council, an arm of the National Academy of Sciences, in 2009. The NRC found that "the relative importance of long-range pollutant contributions from foreign sources is likely to increase as nations institute stricter air-quality standards that result in tougher emission controls on domestic sources." The report said that "it is possible that emissions from emerging economies like China, with relatively limited emissions controls, are outpacing reductions in the developed countries." The committee concluded that "long-range transport of pollutants from foreign sources can under some conditions have a significant impact on U.S ambient concentrations and deposition rates for each of the four pollutant classes considered and that U.S. environmental goals are affected to varying degrees by nondomestic sources of these pollutants."

A significant portion of the ozone-related emissions across the country are now dominated by onroad and offroad mobile sources. In fact, for most nonattainment areas, the predominant source of emissions is cars, trucks, and other motor vehicles. Mobile sources are generally outside the direct control of nonattainment areas and states. The reason is that Congress preempted states from adopting "engine standards" on mobile sources under the Clean Air Act, with the well known exception for California. Congress created the preemption clause in recognition of the fact that mobile sources move from state to state and that the most efficient and effective way to address these emissions was through a national program. The manufacturers went along with this scheme because they did not want to face 50 different standards. Because of preemption, however, states and nonattainment areas have very little effective control over the majority of the emission sources in their region. Although states can develop "in-use" restrictions, such restrictions are extremely limited both in terms of their availability as well as their effectiveness. States and nonattainment areas therefore are generally at the mercy of the federal government to control mobile sources.

One need go no further than to consider the different iterations of the interstate transport rule still under development — from CAIR, to CATR, to CSAPR — to understand the impacts of different states' pollution on one another. Emissions from an attainment area in one state can contribute significantly to a designation of nonattainment in another state. Not only can pollution from one state impact an adjoining state, but pollution from one state can impact states half-way across the country. The reason why Texas, for example, was included in the original Clean Air Interstate Rule for particulate matter was a modeled impact on two air monitors in Illinois.

Interstate pollution is outside the direct control of individual states and nonattainment areas.

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One state cannot generally tell another state what to do. Although Congress and EPA provided for a mechanism to recognize other states' pollution impacts (i.e., the petition process in Section 126 of the Clean Air Act), addressing interstate pollutant transport has been difficult. Intrastate pollutant transport can potentially affect pollutant levels in nonattainment areas in the same state. Environmental groups and others have complained about power plants being built just outside nonattainment areas where they can still service the population centers but are not required to comply with more stringent nonattainment controls. They have also complained about growing natural gas development in attainment areas potentially impacting downwind nonattainment areas. Where more sources are being developed in attainment areas, the distinction between attainment and nonattainment of course becomes less and less.

The distinction is also disappearing because of energy advancements. New developments in where and how energy is either created or conserved is making it difficult to geographically link emissions benefits with emissions reductions. EPA, for example, is trying to encourage states to use energy efficiency and alternative energy to address the NAAQS. But when electricity is flowing all over the place and emissions are flowing all over the place, geographically linking the two and then wedding them geographically to a particular nonattainment area becomes an administrative exercise fraught with all kinds of complexities, perils, and costs. And what is the true environmental benefit of this bean allocation exercise? As EPA has stated, "There can be considerable uncertainty as to where the reduced demand from energy efficiency or displaced energy from a renewable source will actually show up as reduced electrical generation." And as EPA's Kathleen Hogan stated, "Geographical alignment of benefits with reductions, that tends to be kind of an expensive test to demonstrate."

Linking an emission reduction project to an emissions benefit in a particular nonattainment area is clearly a difficult exercise — especially where energy might be coming from outside a nonattainment area or might be coming from distributed energy sources such as solar panels and fuel cells. As energy continues to advance, geographically linking emissions and energy to particular geographic areas will likely become even more difficult and less helpful to advancing air quality goals.

The disappearance in the distinction between attainment and nonattainment is making the chief statutory process for achieving attainment, the State Implementation Plan process, even less effective at improving air quality. Back in 2004, the National Research Council wrote the following: "The SIP process now mandates extensive amounts of local, state, and federal agency time and resources in a legalistic, and often frustrating, proposal and review process, which focuses primarily on compliance with intermediate process steps. This process probably discourages innovation and experimentation at the state and local levels; overtaxes the limited financial and human resources available to the nation's [air quality management] system at the state, local, and federal levels; and draws attention and resources away from the more germane issue of ensuring progress toward the goal of meeting the NAAQS."

As the CAA specifies, "Each state shall have the primary responsibility for assuring air quality within the entire geographic area comprising such state by submitting an implementation plan for such state which will specify the manner in which national primary and secondary ambient air quality standards will be achieved and maintained." When the statute was written 42 years ago and the SIP process was created, emissions from other countries were relatively small, localized emissions were much higher, and pollutant transport was not as much of a concern. The world has since changed. With the disappearing distinction, the SIP process has become an even less effective tool in improving air quality.

Our nation can either continue spending considerable amounts of time and money making these distinctions and trying to work around them — or we can try another route. Franklin Roosevelt once said, "Do something. If it works, do more of it. If it doesn't, do something else." The distinction between attainment and nonattainment worked for many years. The CAA will be subject to a torrent of change in the coming years. Improvements can be made to address the new realities of our world. The alternative is to watch the statute change itself. •

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